Spatio-Temporal Surface Defect Detection via Photorealistic Drone Simulation

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https://github.com/zaherarman/aerial-surface-defect-segmentation

Introduction

Manual human inspection of critical infrastructure, such as bridges and wind turbines, is a high-risk, time-consuming activity prone to human error. While drones can capture high-resolution imagery, the automated detection of surface defects (e.g., micro-cracks or corrosion) is hindered by a lack of diverse, labeled datasets, especially temporal video data. The goal of this project is to develop a deep learning pipeline that utilizes synthetic video data from Unreal Engine 5 (UE5) to train a spatio-temporal model capable of robust defect detection. This project leverages "Sim2Real" transfer learning to bypass data scarcity. Deep learning is the ideal tool here as traditional edge-detection algorithms cannot distinguish between structural cracks and environmental noise (e.g., shadows, bird droppings, or moisture), whereas temporal neural networks can analyze feature persistence across multiple frames to improve confidence and reduce false positives. Ultimately, this work is important because it enables earlier, safer, and more cost-effective detection of structural failures, reducing risk for human workers while improving the reliability and longevity of critical infrastructure.

Illustration

The proposed system consists of a three-stage pipeline, as illustrated in Figure 1: (i.) A UE5 simulation environment utilizing procedural textures and drone physics plugins (e.g., AirSim) to generate annotated video sequences. (ii.) A spatio-temporal feature extractor (CNN-LSTM) that processes frame sequences. (iii.) An output layer providing a classification or probability map of detected defects.

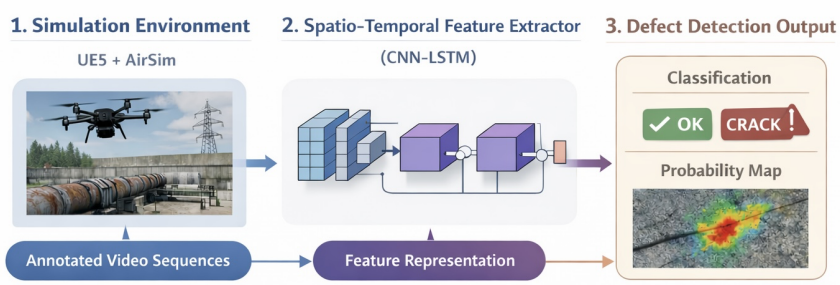

Figure 1: System Pipeline: Synthetic Generation to Spatio-Temporal Inference.

Background & Related Work

To contextualize this project, we build upon five key areas of research:

1. **Simulation Frameworks:** Shah et al. [1] developed AirSim, providing the physical and visual fidelity required for realistic drone kinematics and sensor simulation.
2. **Deep Learning for Inspection:** Cha et al. [2] demonstrated the efficacy of Faster R-CNNs for structural damage detection, establishing a benchmark for CNN-based inspection.

3. **Sequence Learning:** Donahue et al. [3] proposed Long-term Recurrent Convolutional Networks (LRCNs), which combine CNNs with LSTMs for visual time-series tasks.
4. **Procedural Data Generation:** Modern engines like UE5 implement procedural decals to simulate randomized, high-fidelity crack patterns, building on foundational graphics research [4, 5].
5. **Domain Adaptation:** Research in Sim2Real transfer addresses the "reality gap" using domain randomization [6] to ensure synthetic training generalizes to real-world textures.

Data Processing

The dataset will be synthetically generated using Unreal Engine 5 to ensure perfect label accuracy and high environmental variety.

- **Sources:** Will use PBR (Physically Based Rendering) materials for concrete and metal, applying procedural "decals" to simulate randomized cracks and oxidation.
- **Domain Randomization:** Will programmatically vary lighting angles (Global Illumination), weather conditions (rain/fog), and drone "camera jitter" to prevent overfitting to static simulation artifacts.
- **Formatting:** Video streams will be sampled at 15 FPS and converted into tensors of shape (N, T, C, H, W). Frames will be normalized using ImageNet statistics and augmented with temporal-consistent transforms to maintain sequence integrity. 15 FPS was chosen as an adequate FPS to avoid motion blur, however this parameter might need to be adjusted.

Architecture

We will utilize a **CNN-LSTM** (LRCN) architecture, a hybrid model designed for sequential image data. The model consists of two main components:

1. **Spatial Feature Extraction:** A pre-trained MobileNetV2 backbone is employed to extract 2D visual features. To accommodate sequential data in PyTorch, the input tensor of shape (N, T, C, H, W) is reshaped by collapsing the batch and temporal dimensions into a single dimension $(N \times T, C, H, W)$. This allows the backbone to perform independent feature extraction on each frame in the sequence before the features are reshaped back for temporal processing.
2. **Temporal Learning:** A Long Short-Term Memory (LSTM) layer that processes the sequence of feature vectors to capture temporal dependencies and motion cues.

This approach allows the model to distinguish between persistent surface cracks and momentary environmental noise, such as moving shadows.

Baseline Model

Our baseline will be a standard 2D CNN (MobileNetV2) trained on individual static frames from the same UE5 dataset. By comparing the baseline's frame-by-frame accuracy against our CNN-LSTM's sequence-based accuracy, we can quantify the performance gain provided by temporal context, particularly in reducing false positives caused by flickering shadows or lens flares.

Ethical Considerations

While designed for safety, this technology could be misused for unauthorized structural vulnerability assessment of private or government property. Furthermore, if the simulation textures lack diversity (e.g., only "new" concrete), the model might fail to detect defects in older, soot-covered infrastructure, leading to a safety bias where older neighborhoods receive lower-quality automated inspections.

References

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